Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Digital Assignment Lab - 3

Course Name :- Agile Development Process and Devops Lab

Course code :- ISWE406P

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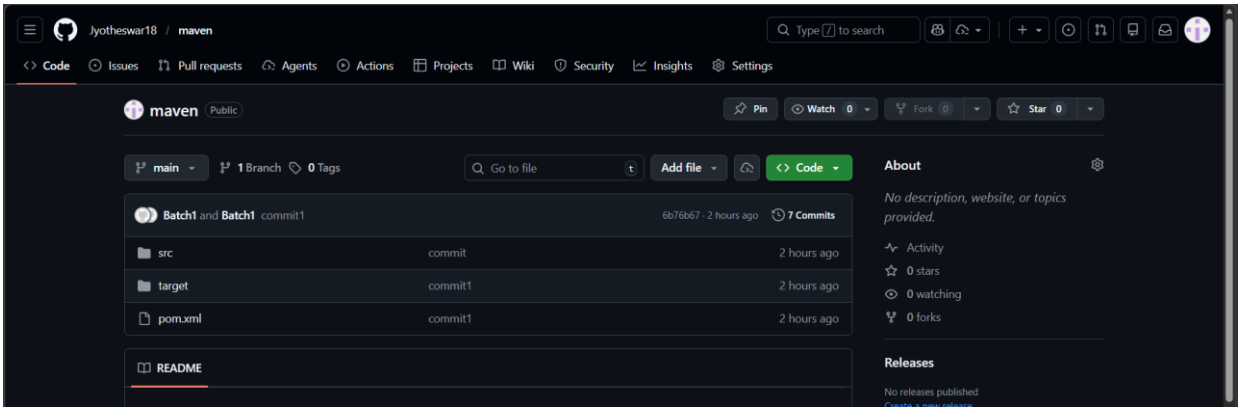
Reg-No:- 23MIS0097

1- Create a Maven Project in Git

- Create a new Git repository



- Create a simple Maven project structure manually, including the following files and folders:

jenkinsfile

pom.xml

src/main/java/com/example/App.java

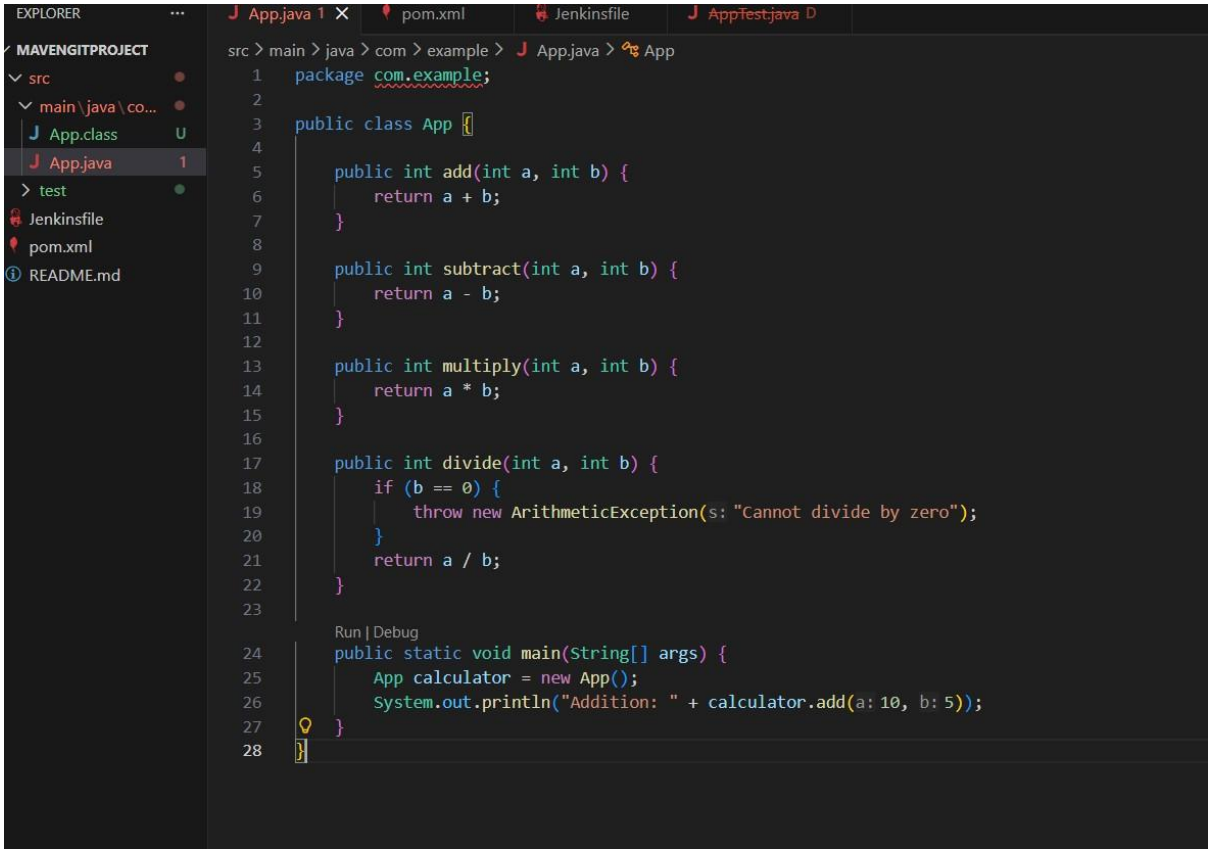
src/test/java/com/example/AppTest.java

```
C:\Users\Batch1\Desktop\cp\maven>tree /F
Folder PATH listing for volume Windows
Volume serial number is 368D-E1FD
C:.
    pom.xml

    src
        main
            java
                com
                    example
                        App.java

        test
            java
                com
                    example
                        Apptest.java
```

2- Implement the Java Program

• Write a simple Java program in App.java under your newly created Git repository for any one of the following scenarios:

Simple Calculator Application

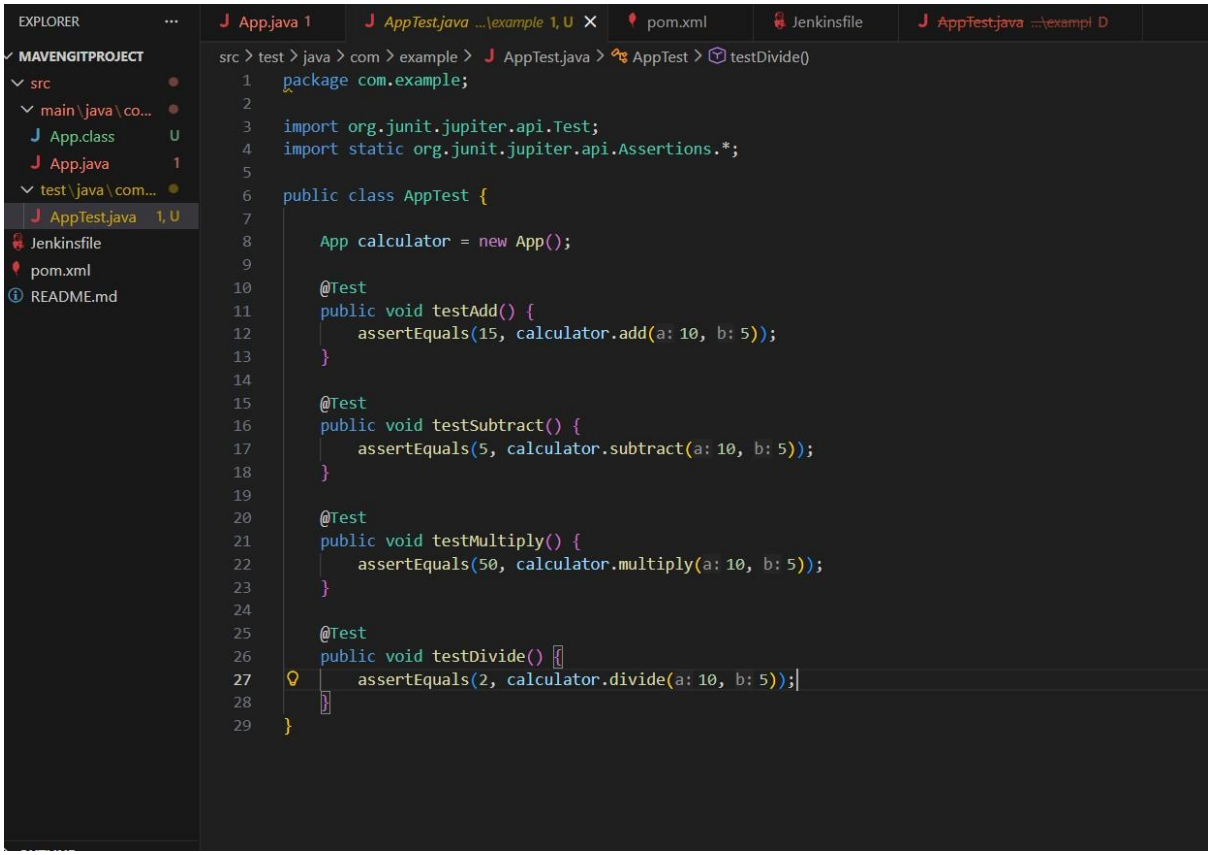


Code :

```
package com.example;

public class App {

    public int add(int a, int b) {

return a + b;

    }


    public int subtract(int a, int b) {

return a - b;
```

```
    }
    public int multiply(int a, int b) {
return a * b;
    }
    public int divide(int a, int b) {
if (b == 0) {
            throw new ArithmeticException("Cannot divide by zero");
        }
        return a / b;
    }
    public static void main(String[] args) {
        App calculator = new App();
        System.out.println("Addition: " + calculator.add(10, 5));
    }
}
```

• Write a corresponding unit test in AppTest.java.

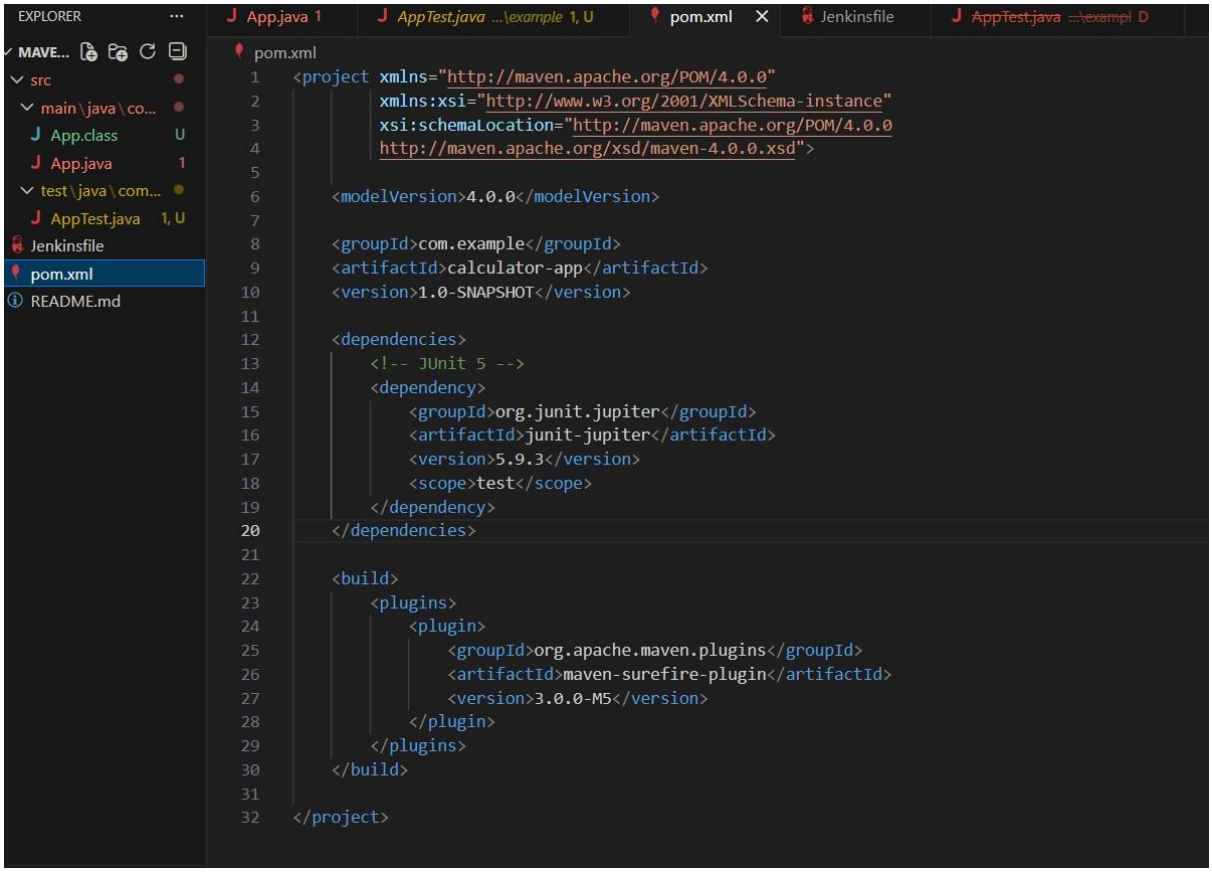


Code :

```
package com.example;
 import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;
public class AppTest {
   App calculator = new App();

   @Test     public void
testAdd() {
       assertEquals(15, calculator.add(10, 5));
   }
   @Test
```

```
    public void testSubtract() {
assertEquals(5, calculator.subtract(10, 5));
    }
    @Test
    public void testMultiply() {
        assertEquals(50, calculator.multiply(10, 5));
    }
    @Test
    public void testDivide() {
assertEquals(2, calculator.divide(10, 5));
    }
}
```

Code for Pom.xml:



3- Push and Clone the Repository

- Clone the repository using Visual Studio Code.

```
sers\Batch1\Dessktop\cp\maven> git add .
ommit -m "Initial Maven Calculator Project"
(root-commit) e43a297] Initial Maven Calculato
anged, 89 insertions(+))
de 100644 pom.xml
de 100644 src/main/java/com/example/App.java
de 100644 src/test/java/com/example/AppTest.java
```

```
PS C:\Users\Batch1\Desktop\cp\maven> git pull origin main --allow-unrelated-histories
 Enumerating objects: 3, done.
 remote: Counting objects: 100% (3/3), done.
 remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
 Unpacking objects: 100% (3/3), 867 bytes | 173.00 KiB/s, done.
 From https://github.com/santhoshh005/MavenGitProject
 * branch        main       -> FETCH_HEAD
 * [new branch]  main       -> origin/main
 Merge made by the 'ort' strategy.
 README.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 README.md

PS C:\Users\Batch1\Desktop\cp\maven> git push -u origin main
 Enumerating objects: 17, done.
 Counting objects:100% (17/17), done.
 Delta compression using up to 16 threads
 Compressing objects: 100% (8/8), done.
 Writing objects: 100% (16/16), 1.82 KiB | 932.00 KiB/s, done.
 Total 16 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
```

- Verify the project structure.

```
C:\Users\Batch1\Desktop\cp\maven>tree /F
Folder PATH listing for volume Windows
Volume serial number is 368D-E1FD
C:.
│   pom.xml
│
├───src
│   ├───main
│   │   └───java
│   │       └───com
│   │           └───example
│   │                   App.java
│   │
│   └───test
│       └───java
│           └───com
│               └───example
│                       Apptest.java
```

```
│   ├───generated-sources
│   │   └───annotations
│   ├───generated-test-sources
│   │   └───test-annotations
│   ├───maven-archiver
│   │       pom.properties
│   │
│   ├───maven-status
│   │   └───maven-compiler-plugin
│   │       ├───compile
│   │       │   └───default-compile
│   │       │           createdFiles.lst
│   │       │           inputFiles.lst
│   │       │
│   │       └───testCompile
│   │           └───default-testCompile
│   │               createdFiles.lst
│   │               inputFiles.lst
│   │
│   ├───surefire-reports
│   │       com.example.AppTest.txt
│   │       TEST-com.example.AppTest.xml
│   │
│   └───test-classes
│       └───com
│           └───example
│                   AppTest.class
```

4- Build the Maven Project

- Use Maven commands in Visual Studio Code terminal to:

Compile the project

Command Used:- mvn compile

```
PS C:\Users\Batch1\Desktop\cp\maven> mvn compile
 Scanning for projects...
[INFO]
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------< com.example:calculator-app >------------------
[INFO] Building calculator-app 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] ----------------------------[ jar ]----------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resource) @ calculator-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is
 platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Batch1\Desktop\cp\maven\src\main\resources
[INFO] ------------------------------------------------------------------------
[INFO] --- compiler:3.13.0:compile (default-compile) @ calculator-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] BUILD SUCCESS ------------------------------------------------------
[INFO] Total time: 0.400 s
[INFO] Finished at: 2026-03-02T21:03:04+05:30
[INFO] ------------------------------------------------------------------------

PS C:\Users\Batch1\Desktop\cp\maven>
```

Run tests

Command Used:- mvn test

```
C:\Users\Batch1\Desktop\cp\maven>mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:simple-maven-app >-------------------
[INFO] Building simple-maven-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Batch1\Desktop\cp\maven\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ simple-maven-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Batch1\Desktop\cp\maven\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ simple-maven-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ simple-maven-app ---
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.592 s
[INFO] Finished at: 2026-03-05T15:21:51+05:30
[INFO] ------------------------------------------------------------------------
```

```
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.03 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.557 s
[INFO] Finished at: 2026-03-02T21:03:10+05:30
[INFO] ------------------------------------------------------------------------
```

Build the project (mvn clean install)

Command Used:- mvn clean install

```
PS C:\Users\Batch1\Desktop\cp\maven> mvn clean install
 Scanning for projects...
[INFO] ------------------< com.example:calculator-app >---------------
[INFO] Building calculator-app 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] ----------------------------[ jar ]----------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/revice/branc5t50/maven-jar-plugin-3.4.1
.1.pom (7.8 kB at 14 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-42.pom
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven-plugins/maven- 32.04 (59 kB at 75 kB/s
1.pom
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven-parent-42.pom
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven-plugins-31.2 (53.494 kB at 2.6·374 kB/s
11.pom (5.8 kB at 75 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven-parent-42.pom
[7.212.lar.1.1 (78 kB at 75 kB/s)
[INFO] --- clean:3.2.0:clean (default-clean) @ calculator-app ---
[INFO] Deleting C:\Users\Batch1\Desktop\cp\maven\GitProject\target
[INFO] -------------------------------------------------------------
[INFO] --- resources:3.3.1:resources (default-resources) @ calculator-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build se com!
[INFO] skip non existing resourceDirectory C:\Users\Batch1\Desktop\cp\maven\GitProject\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:install (default-compile) @ calculator-app ---
[INFO] Installing 1 source file with javac 1.8.0_181 to targetclasses
[WARNING] Consider compiling source at a lower level to be compatible compatile with Java 8
[WARNING] Unable to locate the bootstrop class path to class files that cannot on JDK 8
[WARNING] As a workaround, Use -XlInC- marking option instead because -source 1.8 is taks1.8 because its sets s automatically
[INFO] BUILD SUCCESS ------------------
[INFO] Total time: 5.389 s
[INFO] Finished at: 2026.03.02T21:04:13 +05:30
PS C:\Users\Batch1\Desktop\cp\maven>
```

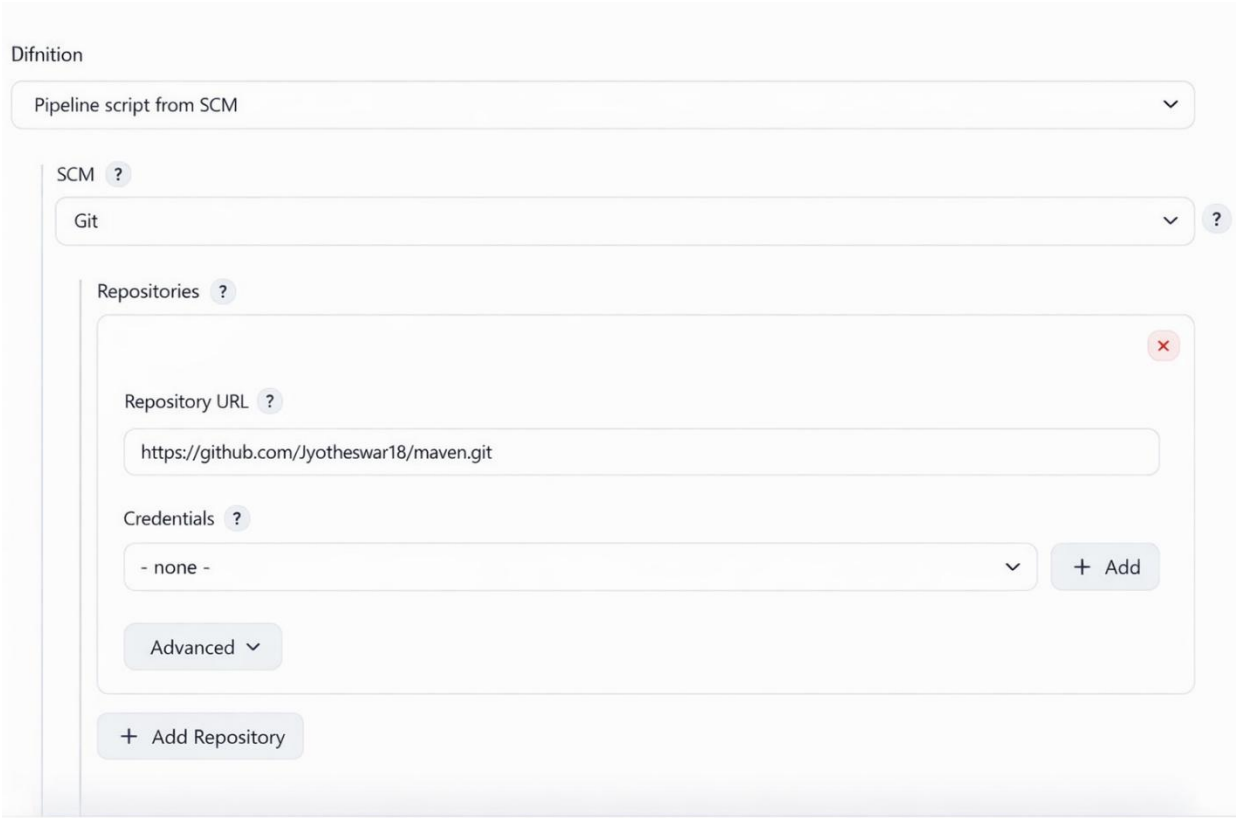
```
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ calculator-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\saisa\OneDrive\Desktop\GitProject\MavenGitProject\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ calculator-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
  not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.0.0-M5:test (default-test) @ calculator-app ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.029 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ calculator-app ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-management/3.1.0/file-management-3.1.0
.pom
```

5- Create and Run Jenkins Pipeline

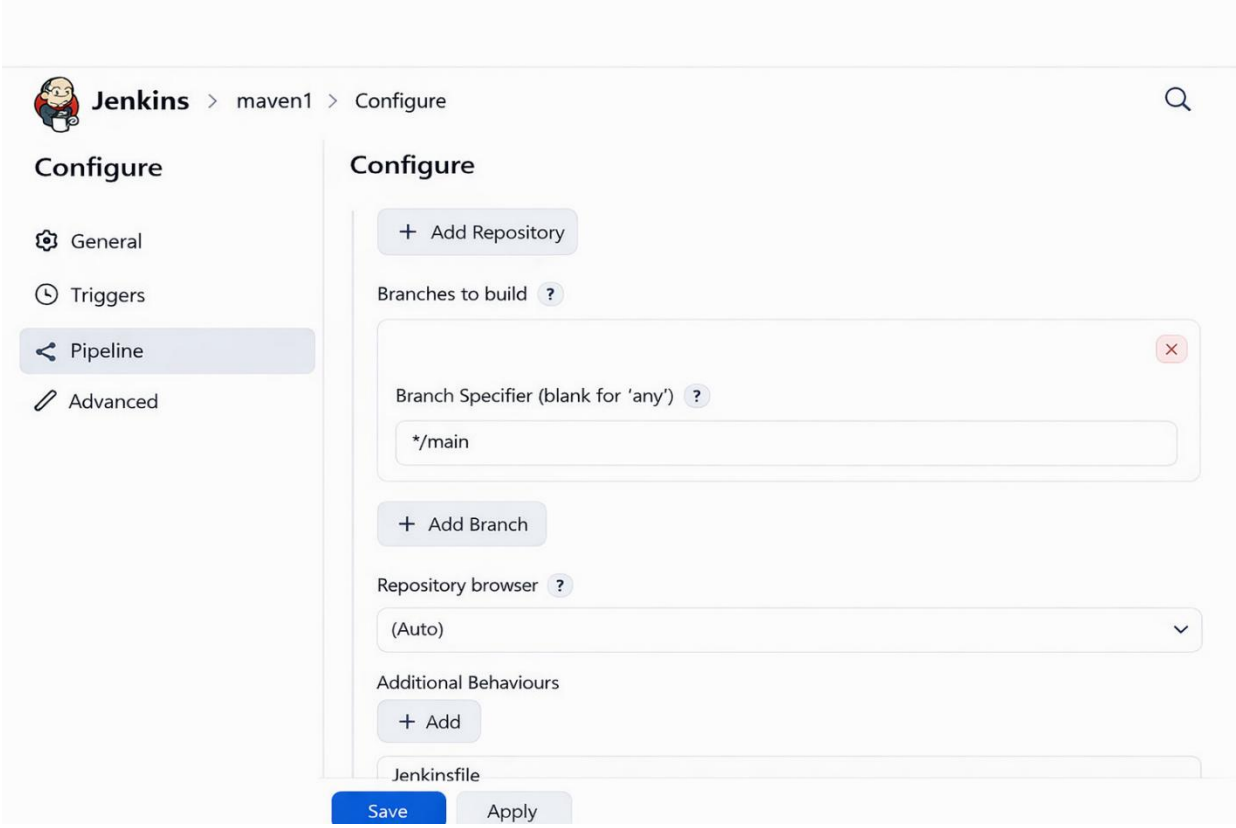
• Create a new Pipeline project in Jenkins.

• Create a pipeline script pipeline {

```
agent any

    stages {
        stage('Clone Repository') {
steps {
                git ‘https://github.com/Jyotheswar18/maven.git'
            }
        }
        stage('Build') {
steps {
                bat 'mvn clean install'
            }
        }
    }
}
```



• Execute the pipeline and verify successful build completion.



Jenkins / maven1

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Full Stage View
- Favorite
- Job Config History
- Open Blue Ocean
- Stages
- Rename
- Pipeline Syntax
- Set Next Build Number

Builds

Filter /

Today

- #3 3:21 PM
- #2 3:02 PM
- #1 2:58 PM

maven1

Stage View

	Declarative: Tool Install	Checkout Git	Build and Test
Average stage times: (full run time: ~12s)	897ms	2s	6s
#3 Mar 05 15:21 — 2 commits	86ms	2s	9s
#2 Mar 05 15:02 — No Changes	97ms	1s	2s failed
#1 Mar 05 14:58 — No Changes	2s	5s	7s failed

Permalinks

- Last build (#3), 6 min 25 sec ago
- Last stable build (#3), 6 min 25 sec ago
- Last successful build (#3), 6 min 25 sec ago
- Last failed build (#2), 25 min ago
- Last unsuccessful build (#2), 25 min ago
- Last completed build (#3), 6 min 25 sec ago